ECSE 331
Laboratory No. 3
MOSFET DC Characteristics in PSPICE

March 16, 2021

Seyed Yazdan Zinati
260928077
seyed.zinati@mail.mcgill.ca

James Wu
260929944
james.wu4@mail.mcgill.ca

***Abstract*—A MOSFET is a three-terminal device that allows current to flow through it when a certain threshold voltage is reached. The three terminals of a MOSFET are called the gate, the source, and the drain. There are two types of MOSFETs: NMOS FET and PMOS FET. Current flows from the drain to the source for NMOS FET and vice versa for PMOS FET. For NMOS FET, current will not flow through the device unless the voltage across the gate terminals and the source terminal V_{GS} is greater than a certain threshold voltage V_t. For PMOS FET, current will not flow through the device unless V_{GS} is less than V_t. MOSFETs have two modes of operation: triode and saturation. The difference between these two modes is that the equation to calculate the current flowing through the MOSFET I_D is different. In triode mode, I_D increases much more rapidly, as the voltage across the drain and source increases V_{DS}, than in saturation mode. NMOS FETs operate in triode mode when $V_{DS} < V_{GS} - V_t$, and they operate in saturation mode when $VDS > V_{GS} - V_t$. For PMOS FETs, the conditions of the operation modes are the opposite of NMOS FET; triode mode when $V_{DS} > V_{GS} - V_t$ and saturation mode when $V_{DS} < V_{GS} - V_t$.**

I. Introduction

MOSFETs are three-terminal devices that allows current to flow through it when V_{GS} is either greater than or less than V_t. The operating mode of the MOSFET depends on V_{DS} compared to $V_{GS} - V_t$.

In this laboratory, we conduct experiments on a circuit that has an NMOS FET using circuit simulation software. PSPICE software is used to build and simulate circuits. The circuit that will be built using PSPICE is shown in Figure 1. We first examine the $I_D - V_{DS}$ characteristics of the MOSFET. Then, we determine the MOSFET transconductance. Finally, we examine the effects of temperature on MOSFET.

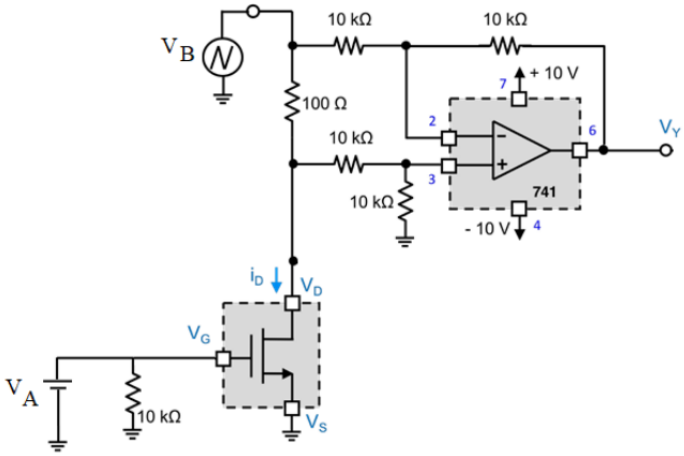


Fig. 1. Setup for measuring MOSFET $I - V$ characteristics

A. Task 1

The first part of the lab involves measuring NMOS FET characteristics. First, we plot V_{DS} vs time and I_D vs time. It should be observed that I_D remains zero despite any change in V_{DS}. We then determine the threshold voltage of the NMOS FET by increasing V_{GS} until I_D exceeds 0.5mA.

Next, we plot the voltage output (V_Y) of the op-amp divided by 100 vs time and the current through the 100Ω resistance vs time. These two plots are then compared with each other and examined.

Finally, we increase V_{GS} in increments of 1V until V_{GS} reaches 5V while plotting I_D vs V_{DS} for each value of V_{GS}. From the plot of I_D vs V_{DS} for each V_{GS} value, we can obtain the early voltage of the MOSFET. This can be done by extending the linear trends of the saturation modes of the traces to the left until they reach $I_D = 0$. The corresponding V_{DS} value is the negative value of the early effective voltage.

B. Task 2

The second part of the lab involves the computation of the MOSFET transconductance (g_m) of the NMOS FET. The transconductance is found using the following equation:

$$g_m = \frac{\Delta i_D}{v_{GS}}$$

The circuit must first be modified by replacing the saw-tooth signal V_B with a DC voltage source set to 5V and V_A needs to be set to 3V. I_D and V_GS is recorded at these values and they're recorded again when V_A is adjusted by 0.1V. ΔI_D and ΔV_{GS} are then found by taking the differences between the measured V_{GS} and I_D and the transconductance is computed.

C. Task 3

The third part of the lab examines the effect of temperature on the MOSFET. First, the circuit must be restored back to its original state in Figure1 by using the saw-tooth waveform as V_B and setting V_A to 3V. The simulation is conducted at temperatures 10°C, 20°C, 27°C, 35°C and 50°C. The I_D vs V_{DS} plots at these temperatures are compared.

II. Analysis

A. Task 1

Task 1 of the experiment involves examining the I-V characteristics of the MOSFET.

We first construct the circuit shown in Figure1 in PSPICE using a uA741 op-amp, a MbreakN3 NMOS FET with $W = 10\mu m$ and $L = 1\mu m$ and V_B as a saw-tooth waveform over a 0V to +10V signal range using the VPULSE component with the following characteristics:

$V1 = 0$
$V2 = 10$
$TD = 0$
$TR = 0.49e^{-3}$
$TF = 0.49e^{-3}$
$PW = 1e^{-5}$
$PER = 1e^{-3}$

We implement the -10V and +10V voltages on pin 4 and pin 7 of the uA741 op-amp using two DC voltage sources. The constructed schematic in PSPICE is shown in Figure 2.

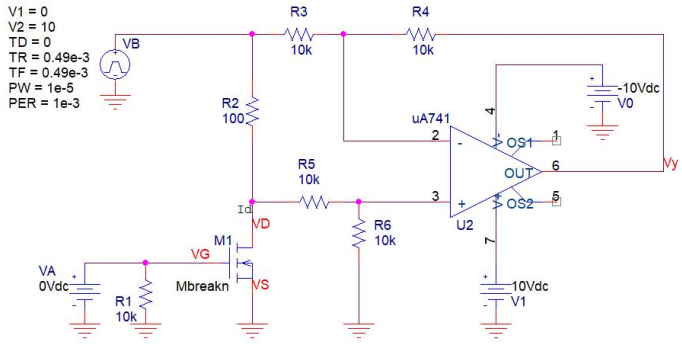


Fig. 2. Task1 circuit number 1

Initially, V_A is set to 0V. When the circuit is constructed, we perform a time domain simulation with $V_A = 0$ to determine to obtain plots of V_{DS} vs time and I_D vs time. The plots of V_{DS} vs time and I_D vs time can be seen in Figure 3 and Figure 4 respectively.

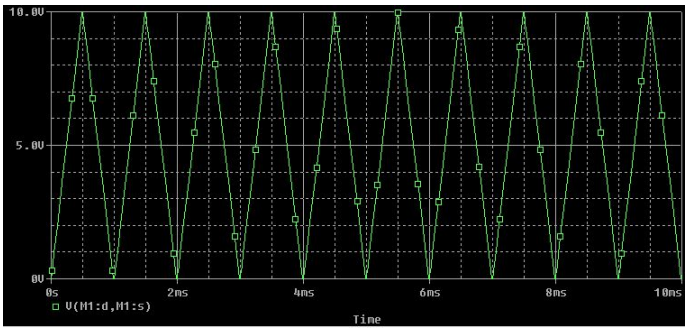


Fig. 3. V_{DS} vs time with $V_A = 0V$

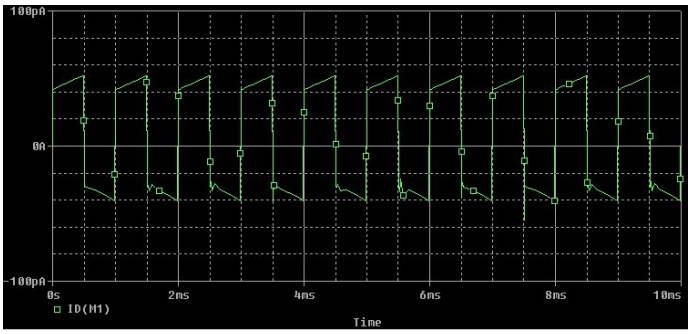


Fig. 4. I_D vs time with $V_A = 0V$

It can be seen from Figure 3 that the voltage V_{DS} follows a triangular waveform over a 0V to 10V signal range with a period of 1ms. On the other hand, the FET drain current simulation shown in Figure 4, is in the range of picoamperes which is close to 0. This is expected as the gate voltage V_G is equal to zero. Therefore, the drain current I_D remains negligible irrespective of any change in the drain-source voltage V_{DS}.

Next, to determine the threshold voltage V_t, V_{GS} is increased in 1V increments until I_D becomes greater than 0.5mA. Once I_D becomes greater than 0.5mA, the voltage V_{GS} is considered to be the threshold voltage V_t. To increase V_{GS} in 1V increments, V_A is increased in 1V increments as the value of V_A is equal to the value of V_{GS}.

The value of V_A is increased from 0V to 1V and the corresponding I_D vs time plot is obtained. At $V_{GS} = 1V$, the I_D vs time plot can be seen in Figure 5. From Figure 5, we can see that I_D isn't greater than 0.5mA, thus we need to continue increasing V_{GS} by 1V.

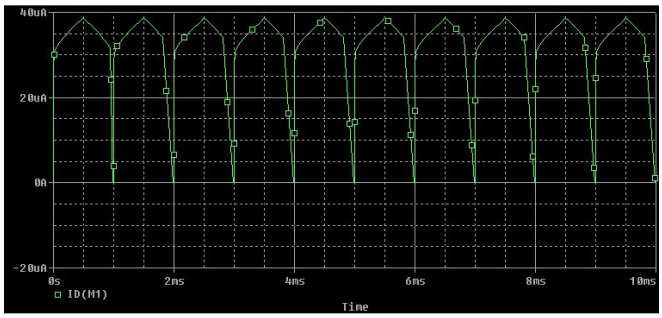


Fig. 5. I_D vs time with $V_A = 1V$

After the value of V_A is increased from 1V to 2V, its corresponding I_D vs time plot is obtained. The plot of I_D vs time can be seen in Figure 6. Based on the plot, I_D is found to be approximately 0.65mA which is greater than 0.5mA. Thus, the threshold voltage V_t is determined to be 2V.

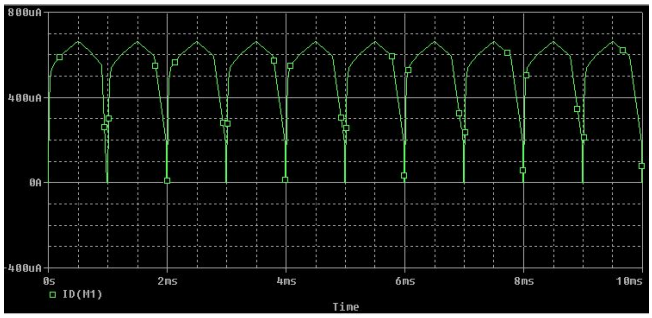


Fig. 6. I_D vs time with $V_A = 2V$

B. Task 2

C. Task 3

III. Conclusion

IV. Appendix